

--	--	--	--	--	--	--	--

B.Tech. Degree VII Semester Special Supplementary Examination May 2017

ME 1703 MACHINE DESIGN II (2012 Scheme)

(Design Data books are permitted)

Time : 3 Hours

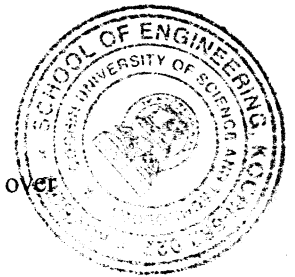
Maximum Marks : 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I.
 - (a) Explain the working of a centrifugal clutch with a neat sketch.
 - (b) How are brakes rated and classified?
 - (c) What is meant by interference in gears? Explain.
 - (d) Explain with neat sketch the spur gear nomenclature.
 - (e) Discuss the advantages and disadvantages of ball and roller bearings over journal bearings.
 - (f) What are the desirable properties of bearing materials?
 - (g) What are the stresses developed in a welded joint and how are they relieved?
 - (h) What are the main design considerations for casting?



PART B

(4 × 15 = 60)

- II. A stone crushing machine receives power from a motor rated at 50 kW at 1800rpm by means of flat belts. The pulley diameters are 200 mm and 700 mm. Centre distance between two pulleys is 4000 mm. Design the belt drive if the direction of rotation of the two pulleys are opposite to each other.

OR

- III. Design a chain drive to actuate a compressor from a 10 kW electric motor at 960 rpm. The compressor speed is to be 350 rpm. Minimum center distance should be 0.5 m. Motor is mounted on an auxiliary bed. Compressor is to work for 8 hours/day.

- IV. Design a spur gear to transmit 22 kW at 900 rpm with a speed reduction of 2.5. Materials for pinion and wheel are C15 steel and cast iron grade 30 respectively. Take pressure angle of 20° and working life of the gears as 10,000 hours.

OR

- V. Design a bevel gear drive to transmit 7 kW at 1600 rpm for the following data.

Gear ratio	= 3
Material for pinion and gear	= C 45 steel
Life	= 10,000 hours.

(P.T.O.)

VI. Design a journal bearing for a generator to the following specification.

Load on the journal = 12000 N
Diameter of the journal = 75 mm
Speed of the journal = 1400 rpm.

OR

VII. The radial reaction on a bearing is 9000 N. It also carries a thrust of 5000 N. The speed of the shaft is 1000 rpm. The outer ring is stationary. Expected average life of bearing is about 25,000 hours. The load on the bearing is smooth; the service is 8 hours/day.

- (a) Select a suitable roller bearing.
- (b) What is the rated 90% life of selected bearing?
- (c) Compute the probability of the selected bearing surviving 25,000 hours.

VIII. What are the important points to be considered while designing parts to be manufactured by forging? Explain with sketches.

OR

IX. Explain in detail the steps involved in the preparation of work drawings for manufacture of parts.
